

Giridhar Reddy Karnati

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RF and microwave instrumentation · wide-band antenna design and EM simulation · radio receiver characterisation · field-deployable ground stations for satellite telemetry.

EDUCATION

Sep 2022 – Jul 2026	B.E., Electronics & Communication Engineering; Minor in Computer Science Chaitanya Bharathi Institute of Technology (CBIT), Hyderabad ◦ CGPA 9.23 / 10 (as of May 2026) ◦ Academic Excellence Award (Silver Medal) for the highest CGPA up to Semester II (AY 2022–23) in B.E. (ECE-1) ◦ Coursework: Antennas & Wave Propagation, Microwave Engineering, RF & Microwave Circuits, Electromagnetic Theory, Communication Systems, Digital Signal Processing, Digital System Design (Verilog) ◦ Major project: <i>Design of Anti-Spoofing Antenna for Electronic Warfare</i> (guide: Dr. K. Suman)
2020 – 2022	Intermediate (Class XI–XII), Mathematics–Physics–Chemistry Sri Chaitanya Junior Kalasala, Hyderabad ◦ Aggregate: 96.2%
2020	Class X VijayaRatna High School

RESEARCH & PROFESSIONAL EXPERIENCE

Jul 2026 – present	NPTEL Pre-Doc Fellow (Project Associate) IC Design Lab, Dept. of Electrical Engineering, IIT Hyderabad · Advisor: Prof. Abhishek Kumar Front-end for a wideband receiver in CMOS.
May 2026 – Jul 2026	Facility Technologist Intern National Nanofabrication Centre (NNfC), CeNSE, IISc Bengaluru
Nov 2025 – May 2026	Visiting Research Student Astronomy & Astrophysics, Raman Research Institute (RRI), Bengaluru Architectural design and bring-up of a water-vapour radiometer; RF link-budget and thermal-noise analysis, and gain / noise-figure cascade modelling of the IF chain.
Aug 2025 – Sep 2025	Student Intern, Nanofabrication Lab (SPARC NANO-X) IIT Hyderabad Clean-room processing: laser lithography, sputtering, reactive-ion etching (RIE) and plasma cleaning. Trained on the IC design flow from layout through packaging; fabricated nano-silicon waveguides, microfluidic devices and 50 μm Au interdigitated electrodes for 2-D-material gas sensors using Heidelberg μPG 501 maskless lithography, DC magnetron sputtering and lift-off in a Class-1000 clean room.
Jul 2025 – Aug 2025	IEEE AP/MTT-S Summer Intern, RF & Microwave Circuits IIT Kharagpur (remote) · Mentor: Shrawan Kumar Patel RF and microwave circuit analysis and synthesis: designed and simulated microwave phase shifters and coupled-line couplers in Keysight ADS, with MATLAB models for comparison; evaluated insertion loss, phase error and return loss across the band.
Jul 2025	Electrical Engineering Summer School (EESS-25) Indian Institute of Science (IISc), Bengaluru Selected as one of 100 students nationwide for an intensive programme in signal processing, communications and microelectronics.

Jun 2025 – Jul 2025	Project Intern, SDR & Embedded Radio Systems National Institute of Technology Karnataka (NITK), Surathkal Built SDR-based transceivers and Raspberry Pi-driven portable APRS ground nodes for weather monitoring, deployable to remote field sites.
Mar 2025 – May 2026	Internship Trainee, Antenna Design & RF Hardware (part-time) RaoS Consultants LLC (remote) Comprehensive Internship Course on Antenna Design (CICAD 2025–26): electromagnetics, antenna theory, horns, reflectors and phased arrays, design and analysis, integration, and measurement. Tools: CST Microwave Studio. One of 24 students selected from 90+ applicants; 2nd Place Best Intern.
Dec 2024 – Jan 2025	Technical Intern India Space Week (remote)
Oct 2024	Research Trainee, ICAN 4.0 Workshop CREST, BITS Pilani Hyderabad Campus Fabricated and electrically characterised 2-D-material-based nano-device sensors.
Jun 2024 – Jul 2024	Internship Trainee National Institute of Electronics & Information Technology (NIELIT)

PUBLICATIONS

2025	VayuVani: A Self-Managed LoRa Ground Station for Amateur Operators K. Giridhar Reddy · HAM Fest India 2025 Magazine, Goa, pp. 20–27
2025	Design of Ground Station Antennas for Tracking of LoRa-Based Satellites at 433 MHz Amateur Frequencies K. Vasanth, K. Giridhar Reddy, V. Bala Sai Datta, K. Suman · Proc. IEEE SCEECs 2025, MANIT Bhopal, January 2025 · IEEE Xplore Best Paper Award
2024	Privacy-Preserving LoRa Ground Stations in Amateur Satellite Communications: A Case Study of VayuVani K. Giridhar Reddy · Radio Amateurs' World (RAW) Magazine · presented at HAM Fest India 2024, Kolkata

VayuVani is an open-source LoRa ground-station platform created by Chandra Tungathurthi (vayuvani.com). My articles and talks describe building and operating ground stations with it.

HONOURS & AWARDS

Aug 2026	IEEE SPACE Young Professional (YP) Laureate IEEE SPACE 2026 (Space, Aerospace and Defence Conference)
Jun 2026	Best Intern Award, 2nd Place CICAD 2025–26, RaoS Consultants LLC
2025 & 2026	GATE qualified, Electronics & Communication Engineering GATE 2025 (score 436) and GATE 2026
Jan 2026	Dayton Amateur Radio Association Scholarship ARRL Foundation, USA · featured in <i>QST</i> , ARRL's monthly journal
Dec 2025	B.Tech Initiative Travel Grant IEEE Microwaves, Antennas and Propagation Conference (MAPCON)
Dec 2025	Rank 20 of 250 (top 8%), AI/ML Powered FPGA Design & Simulation Hackathon National Institute of Technology Jamshedpur
Jul 2025	Selected for the IISc Electrical Engineering Summer School (EESS-25) Top 100 nationwide

Mar 2025	1st Place, Paper Presentation, VANSH-2K25 Vignan Institute of Technology and Science, Deshmukhi, Hyderabad
Mar 2025	1st Place, Poster Presentation (Pradarshana), Aakruthi 2025 National Technical Symposium, Dept. of ECE, University College of Engineering, Osmania University, Hyderabad
Jan 2025	Best Paper Award (ECE) IEEE SCEECs 2025, MANIT (NIT) Bhopal, 18–19 January 2025
Jan 2025	2nd Place, Quiz Competition, NAFED08 VSSC / ISRO, held at IIT Hyderabad
Mar 2024	Academic Excellence Award (Silver Medal) CBIT, Institute Day (SHRUTHI-2024) · highest CGPA up to Semester II (AY 2022–23) in B.E. (ECE-1)
2024	Three-time NPTEL Star Discipline Star (Computer Science & Engineering) and Believer Star, Jan–Apr 2024 · Enthusiast Star, Jul–Dec 2024
TALKS	
Oct 2025	VayuVani: your private LoRa satellite ground station HamFest India 2025, Goa College of Engineering, Farmagudi, Goa
Dec 2024	Privacy-preserving LoRa ground stations: a case study of VayuVani HAM Fest India 2024, Kolkata
Dec 2024	Speaker, 7th Lamakaan Annual Radio Convention Lamakaan Amateur Radio Club, with the Dept. of ECE & IEEE, Muffakham Jah College of Engineering & Technology, and the Digital Empowerment Foundation, Hyderabad
LEADERSHIP & OUTREACH	
Oct 2023 – Apr 2026	Founder & Student President, then Senior Student Advisor HAM Radio Club of CBIT, Hyderabad - Founder & Student President, Oct 2023 – Nov 2025; Senior Student Advisor, Oct 2025 – Apr 2026. - Led technical projects and STEM-outreach workshops for school students. - Two-day workshop with the Dept. of ECE, “Design approaches of Antennas for HAM Radio Applications using Open Source Software” (Apr 2024): antenna modelling in 4NEC2, Morse code, cables and connectors, and licensing. - Hands-on HAM radio sessions for Class 10 students at ZPHS Nadergul.
2025	Technical Chair & Coordinator, SYNAPSE 2025 CBIT · managed all technical events for the college techno-cultural festival
PROFESSIONAL MEMBERSHIPS	
Jun 2025 – present	Member Applied Computational Electromagnetics Society (ACES)
May 2025 – present	Individual Member RISC-V International
Jan 2025 – present	Associate Member NoRCEL Institute (Network of Researchers on the Chemical Emergence of Life)
Nov 2024 – present	Student Member Institute of Electrical and Electronics Engineers (IEEE)
Sep 2024 – present	Student Member Radio Amateur Satellite Corporation (AMSAT)

Aug 2023 – Apr 2026 **ISF Member (IETE Students' Forum)**
Institution of Electronics and Telecommunication Engineers (IETE)

LICENCES & CERTIFICATIONS

Feb 2024 **Amateur Wireless Station Licence, General Grade (callsign VU2TIO)**
Department of Telecommunications, Government of India

Sep 2026 **IEEE MTT-S Virtual Workshop on AI for Microwaves**
IEEE Microwave Theory and Technology Society

Nov 2025, May 2026 **IEEE Certificates Program (CEU/PDH), two certificates**
IEEE Educational Activities - topics include generative AI and 5G

Mar 2026 **International Physics Competition (IPhyc) 2025, semi-final round**
Participation certificate

Dec 2025 **International Youth Math Challenge (IYMC) 2025, qualification round**
Participation certificate

Feb 2024 **Certificate of Merit, Geo-data Sharing and Cyber Security**
Indian Institute of Remote Sensing (IIRS), ISRO

TECHNICAL SKILLS

EM & RF simulation	Ansys HFSS, CST Studio Suite, Altair FEKO, Keysight ADS, MATLAB RF Toolbox
RF & microwave	Wide-band antennas, LNA, mixer, PLL/VCO, power amplifier, filter design, link-budget analysis, noise figure / IIP3, anechoic-chamber and VNA measurement workflows
RF hardware & bench	VNA, spectrum analyser, SDR (RTL-SDR, HackRF), soldering and RF connectorisation, PCB design and prototyping
Digital, VLSI & EDA	Cadence Virtuoso, Xilinx Vivado, Silvaco TCAD (Victory Process / Device / Extract), CleWin, SolidWorks
Programming & HDL	Python, C/C++, MATLAB, Verilog, VHDL, LabVIEW
Embedded	Arduino, Raspberry Pi, ESP32 (Xtensa LX6), RISC-V, Arm Cortex-M, FPGA prototyping
Fabrication	Photolithography, maskless laser writing, DC magnetron sputtering, reactive-ion etching (RIE), plasma cleaning, PVD, CVD, lift-off, RTA; I–V characterisation (Keithley 2450)
Languages	English (professional), Hindi (native / bilingual)